

Pedro da Costa Abreu Junior

Brasilia, Brazil · pedro@typetheoryforall.com · pedroabreu0.github.io · github.com/pedrotst · LinkedIn

Research Profile

Computer scientist with research experience in proof assistants, programming languages, mechanized metatheory, and formal verification. My work has focused on Coq/Rocq, Isabelle, SAW, Tamarin, ProVerif, and related tools for specifying and reasoning about programming-language and systems artifacts. My current research interests are in the comparative strength and expressive power of formal systems, especially type theory, higher-order logic, set theory, and their mechanization in proof assistants.

Education

M.S. in Computer Science, Purdue University 2024

West Lafayette, IN, USA

Advisor: Prof. Benjamin Delaware. Thesis: *Translating GADTs from OCaml to Coq*.

B.Sc. in Computer Science, University of Brasilia 2017

Brasilia, Brazil

Honours project advised by Prof. Rodrigo Bonifacio: specified and proved correctness properties of Feature Featherweight Java in Coq.

Publications

Pedro Abreu, Benjamin Delaware, Alex Hubers, Christa Jenkins, J. Garrett Morris, and Aaron Stump
POPL 2023

A Type-Based Approach to Divide-and-Conquer Recursion in Coq. ACM SIGPLAN Symposium on Principles of Programming Languages, 2023. doi.org/10.1145/3571196

Current Research

Independent Research: Lean Formalization of Tarski's Undefinability Theorem 2024–present

- Formalizing syntax, parsing, Godel coding, diagonalization, definability, and truth predicates in Lean.
- Using the project to consolidate background in model theory, arithmetic definability, and mechanized foundations.
- Developing supporting parser/unparser infrastructure and determinism proofs for formalized arithmetic syntax.

Research and Professional Experience

R&D Consultant, Free & Fair Winter 2024

Remote

- Contributed to specification and formalization work for end-to-end verifiable internet voting.
- Used tools including Tamarin, ProVerif, Lando, and Clafer to refine protocol and system specifications.

Consultant, Pruvendo Winter 2023

Remote

- Specified the communication mechanism between multiple smart contracts for the EverScale blockchain platform.
- Developed formal definitions in Coq for reasoning about distributed contract interaction.

Research Intern, coq-of-ocaml, Nomadic Labs Summer 2020

Paris / Remote

- Worked on translating OCaml GADTs into Coq within the coq-of-ocaml ecosystem.
- This project became the basis of my M.S. thesis.

Research Intern, Galois / Amazon RTI / ElectionGuard Team Summer 2019

Portland, OR, USA

- Specified and proved safety properties of real-world software using SAW-Script.
- Contributed Coq specifications related to voting-protocol verification.

Research Intern, Kami Team, SiFive 2018

San Mateo, CA, USA

- Worked on proving correctness properties of a RISC-V floating-point unit using Coq and Kami.

Software Engineering Intern, Tribunal de Contas da Uniao 2015–2017
Brasilia, Brazil

- Developed and maintained integration, end-to-end, component, and performance tests for institutional software systems.

Research Intern, Trustworthy Systems, NICTA Summer 2014
Sydney, Australia

- Worked on file-system verification using Isabelle.

Undergraduate Researcher, CIC/UnB Formal Reasoning Group 2013–2014
Brasilia, Brazil

- Researched extensions to Coq rewrite tactics for a formalization of the lambda calculus with explicit substitutions.

Teaching, Mentorship, and Awards

ACM Graduate Teaching Assistant Award 2019
Purdue University

Awarded for teaching contributions as a graduate teaching assistant.

CS565/464: Graduate/Undergraduate Programming Languages Teaching Assistant Fall 2021, Spring 2022, Spring 2024
Purdue University

Supported graduate-level and undergraduate-level instruction in programming languages.

Mentor, SIGPLAN-M 2022–present

- Mentored five students and early-career researchers through SIGPLAN-M.
- Focused on academic planning, advisor relationships, work-life balance, and research direction.

Head Teaching Assistant, Introduction to Computer Science 2017
University of Brasilia

- Managed the selection and coordination of a 36-person teaching-assistant team.

Selected Projects, Outreach, and Service

Founder and Host, Type Theory Forall Podcast 2020–present

- Founded and host a podcast making type theory, programming languages, proof assistants, and formal methods accessible to a broader technical audience.
- Produced over 90 hours of interviews with researchers including Lawrence Paulson, Thorsten Altenkirch, Conal Elliott, and Leonardo de Moura.

Founder and Lead Mentor, TTFA Mentorship Program 2025–present

- Provide structured one-on-one mentoring on research direction, type theory and programming-languages preparation, proof-assistant study, and graduate-school applications.

Coquedille 2019–2020

- Explored translation from the intrinsic type theory of Coq to the extrinsic type theory of Cedille using Coq and MetaCoq.
- Repository: github.com/pedrotst/coquedille

Java Virtual Machine 2016
University of Brasilia

- Led a team of five students in developing a working JVM for bytecode generated by `javac 5`.
- Repository: github.com/pedrotst/JVM

Technical Skills

Proof assistants and formal methods: Coq/Rocq, Lean, Isabelle, Cedille, MetaCoq, Tamarin, ProVerif, SAW, Alloy, Dafny, Clafer. **Programming languages:** OCaml, Haskell, Rust, C/C++, Java, Python. **Other technical background:** Java bytecode, LLVM, formal specification, mechanized metatheory, software verification.

Languages

Portuguese: native. **English:** fluent. **Spanish:** intermediate understanding, basic speaking. **French:** basic. **Esperanto:** advanced.